National University of Singapore

Bridging Mathematics

Semester 1, AY2025/2026

Time allowed: 1 hour and 30 minutes

Mock Test
Sample Solutions

Question 1 [10 marks]

Determine whether each of the following statements are true or false. Justify your answers.

(a) There exists a linear system with exactly three solutions.

False. Every linear system has either no solution, a unique solution, or infinitely many solutions.

(b) If $\mathbf{A}$ is a symmetric matrix, then $\mathbf{A}^2$ must be a symmetric matrix.

True. Suppose that $\mathbf{A}$ is symmetric, that is, $\mathbf{A}^T = \mathbf{A}$. Then,

$$\left(\mathbf{A}^2\right)^T = (\mathbf{A}\mathbf{A})^T = \mathbf{A}^T\mathbf{A}^T = \mathbf{A}\mathbf{A} = \mathbf{A}^2,$$

and $\mathbf{A}^2$ must be symmetric as well.

(c) If $\mathbf{A}$ is an invertible $n \times n$ matrix, then the homogeneous system $\mathbf{A}\vec{x} = \vec{0}$ has infinitely many solutions.

False. $\mathbf{A}$ is invertible if and only if $\vec{y} = \mathbf{A}\vec{x}$ has a unique solution for all $\vec{y} \in \mathbb{R}^n$. Thus, $\mathbf{A}\vec{x} = \vec{0}$ must have a unique solution: i.e., $\vec{x} = \vec{0}$.

(d) The function $T : \mathbb{R}^n \to \mathbb{R}^n$ given by $T(\vec{x}) = \vec{x} + \vec{b}$ is a linear transformation for all $\vec{b} \in \mathbb{R}^n$.

False. Observe that

$$T\left(\vec{0}\right) = \vec{0} + \vec{b} = \vec{b}.$$

If $\vec{b} \neq \vec{0}$, then $T\left(\vec{0}\right) \neq \vec{0}$ and T is not a linear transformation.

(e) You flip a fair coin 100 times. The probability of obtaining exactly 50 heads is greater than the probability of obtaining exactly 25 heads.

True. The number of heads is a binomially distributed random variable with $n = 100$ and $p = 0.5$. Then,

$$P(X = 25) = \binom{100}{25}(0.5)^{25}(0.5)^{75} = \binom{100}{25}(0.5)^{100},$$
$$P(X = 50) = \binom{100}{50}(0.5)^{50}(0.5)^{50} = \binom{100}{50}(0.5)^{100}.$$

Indeed, $P(X = 50) > P(X = 25)$.

Question 2 [10 marks]

In his *Mathematical Classic* (張邱建算經), the 5th Century Chinese mathematician Zhang Qiujian poses the famous *Hundred Fowl Problem*:

> 今有雞翁一直錢五；雞母一直錢三，雞雛三直錢一。凡百錢買雞百隻。問雞翁、 母、雛各幾何?
>
> A rooster costs five coins, a hen costs three coins, and three chicks cost one coin. With 100 coins, we buy 100 of them. How many roosters, hens, and chicks can we buy?

Let r, h, and c be the number of roosters, hens, and chicks respectively. Since we buy 100 birds and spend 100 coins, we have the linear system

$$\begin{cases} r + h + c = 100 \\ 5r + 3h + \frac{1}{3}c = 100 \end{cases}.$$

We row reduce the corresponding augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 100 \\ 5 & 3 & 1/3 & 100 \end{array}\right) \xrightarrow{R_2-5R_1} \xrightarrow{-\frac{1}{2}R_2} \xrightarrow{R_1-R_2} \left(\begin{array}{ccc|c} 1 & 0 & -4/3 & -100 \\ 0 & 1 & 7/3 & 200 \end{array}\right).$$

Let $c = t$. Then, a general solution is given by

$$r = -100 + \frac{4}{3}t, \quad h = 200 - \frac{7}{3}t, \quad c = t, \quad t \in \mathbb{R}.$$

Note that since r, h, and c count animals, they must be non-negative integers. We thus require t be a multiple of 3, and that

$$\begin{aligned} -100 + \frac{4}{3}t \geq 0 \quad &\Longrightarrow \quad t \geq 75, \\ 200 - \frac{7}{3}t \geq 0 \quad &\Longrightarrow \quad t \leq 85. \end{aligned}$$

Thus, we have $t \in \{75, 78, 81, 84\}$, corresponding to the following possible solutions:

$$\begin{pmatrix} \text{roosters} \\ \text{hens} \\ \text{chicks} \end{pmatrix} \in \left\{ \begin{pmatrix} 0 \\ 25 \\ 75 \end{pmatrix}, \begin{pmatrix} 4 \\ 18 \\ 78 \end{pmatrix}, \begin{pmatrix} 8 \\ 11 \\ 81 \end{pmatrix}, \begin{pmatrix} 12 \\ 4 \\ 84 \end{pmatrix} \right\}.$$

Question 3 [10 marks]

Find all matrices $\mathbf{X}$ that satisfy the following equation:

$$\mathbf{X}\begin{pmatrix} -1 & -1 & 0 \\ 0 & -1 & 0 \\ 5 & 3 & 2 \\ -1 & -1 & -2 \end{pmatrix} = \begin{pmatrix} 5 & 3 & 2 \\ 0 & -1 & 2 \end{pmatrix}.$$

Let us write the matrix equation as $\mathbf{XA} = \mathbf{B}$. Taking the transpose of both sides, we have

$$(\mathbf{XA})^T = \mathbf{B}^T \quad \Longrightarrow \quad \mathbf{A}^T\mathbf{X}^T = \mathbf{B}^T.$$

We can thus solve for each column $\vec{x}_i$ of $\mathbf{X}^T$, since

$$\mathbf{A}^T\vec{x}_1 = \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix}, \quad \mathbf{A}^T\vec{x}_2 = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}.$$

We row reduce the augmented matrix:

$$\left(\begin{array}{cccc|c|c} -1 & 0 & 5 & -1 & 5 & 0 \\ -1 & -1 & 3 & -1 & 3 & -1 \\ 0 & 0 & 2 & -2 & 2 & 2 \end{array}\right) \xrightarrow[\frac{1}{2}R_3]{R_2-R_1} \xrightarrow[-R_2]{-R_1} \xrightarrow[R_2-2R_3]{R_1+5R_3} \left(\begin{array}{cccc|c|c} 1 & 0 & 0 & -4 & 0 & 5 \\ 0 & 1 & 0 & 2 & 0 & -1 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right).$$

We construct a general solution for each column of $\mathbf{X}^T$:

$$\vec{x}_1 = \begin{pmatrix} 4s \\ -2s \\ 1+s \\ s \end{pmatrix} \text{ for } s \in \mathbb{R}, \quad \vec{x}_2 = \begin{pmatrix} 5+4t \\ -1-2t \\ 1+t \\ t \end{pmatrix} \text{ for } t \in \mathbb{R}.$$

Hence, the matrix $\mathbf{X}$ is given by

$$\mathbf{X} = \begin{pmatrix} 4s & -2s & 1+s & s \\ 5+4t & -1-2t & 1+t & t \end{pmatrix}, \text{ for } s, t \in \mathbb{R}.$$

Question 4 [10 marks]

Consider the following system of linear equations:

$$\begin{cases} x + ay + bz = 1 \\ x + y + bz = 2 \\ x + y + (a-b)z = -1 \end{cases}.$$

Determine the conditions on a and b such that:

(a) the linear system has no solution;

(b) the linear system has a unique solution, and write down the unique solution;

(c) the linear system has a infinitely many solutions in one parameter, and write down this solution;

(d) the linear system has a infinitely many solutions in two parameters, and write down this solution.

We row reduce the augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & a & b & 1 \\ 1 & 1 & b & 2 \\ 1 & 1 & a-b & -1 \end{array}\right) \xrightarrow[R_3-R_1]{R_2-R_1} \xrightarrow{R_3-R_2} \left(\begin{array}{ccc|c} 1 & a & b & 1 \\ 0 & 1-a & 0 & 1 \\ 0 & 0 & a-2b & -3 \end{array}\right).$$

If $a = 1$, then the matrix becomes

$$\left(\begin{array}{ccc|c} 1 & 1 & b & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1-2b & -3 \end{array}\right),$$

and the system has no solution. If $a \neq 1$, then it suffices to consider the final row: the system has no solution when $a = 2b$, and the system has a unique solution when $a \neq 2b$.

(a) The system has no solution when $a = 1$, or when $a \neq 1$ and $b \neq 1/2$

(b) The system has a unique solution when $a \neq 1$ and $a \neq 2b$. In this case, we can construct the unique solution: we have

$$z = -\frac{3}{1-2b}, \quad y = \frac{1}{1-a}.$$

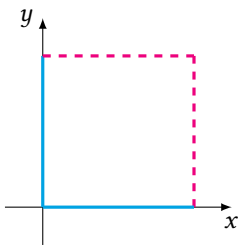
Plugging this into $x = 1 - a - b$, we have

$$x = 1 - a\left(\frac{1}{1-a}\right) - b\left(-\frac{3}{1-2b}\right) = 1 - \frac{a}{1-a} + \frac{3b}{1-2b}.$$

There are no values of a and b for which the linear system has infinitely many solutions in either one or two parameters.

Question 5 [10 marks]

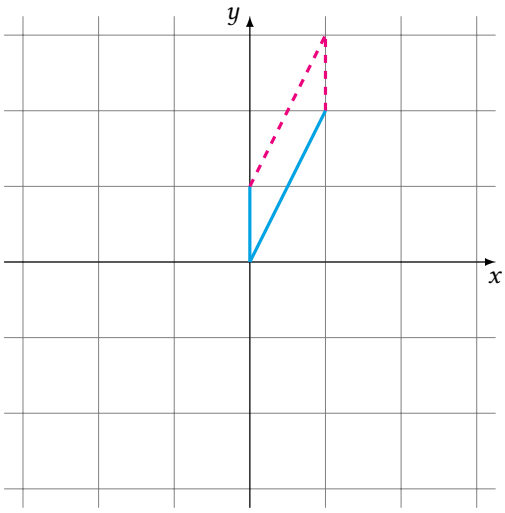
Consider the unit square in $\mathbb{R}^2$ with vertices at $(0,0)$, $(1,0)$, $(0,1)$, and $(1,1)$. Note that the sides along the axes are drawn with filled lines, whereas the remaining edges are drawn with dashed lines.



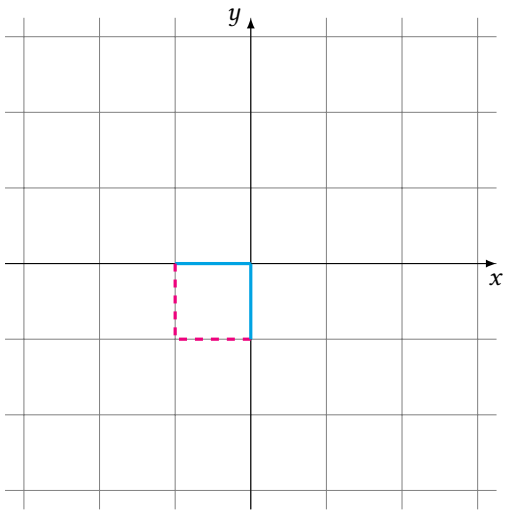
On each set of axes below, draw the image of the square under the linear transformation described by the given matrix.

Show your work to Question 5 in the space below.

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

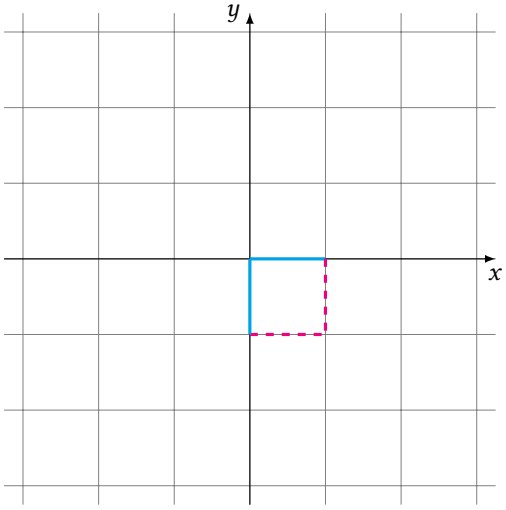


$$\mathbf{B} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

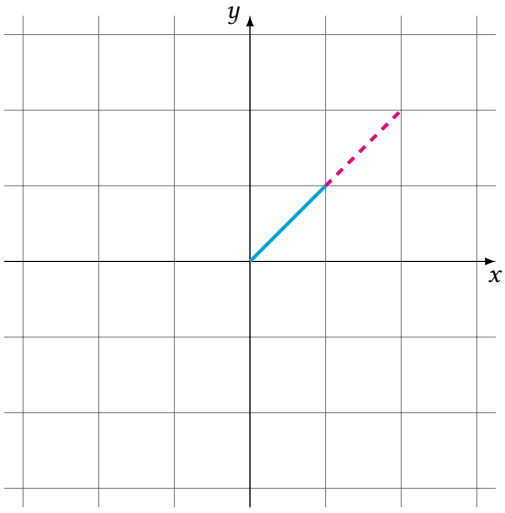


You may use this page to continue your work on Question 5.

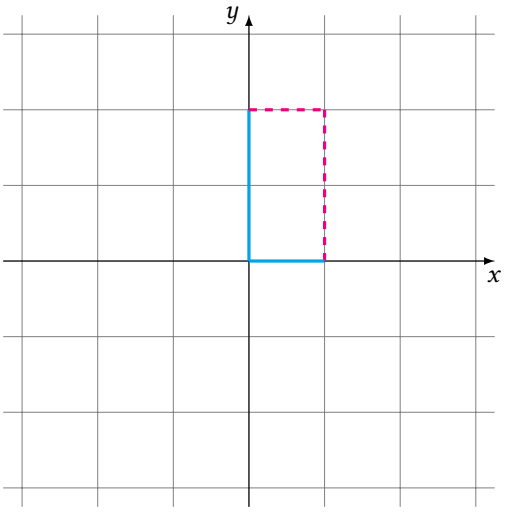
$$\mathbf{C} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$



$$\mathbf{D} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

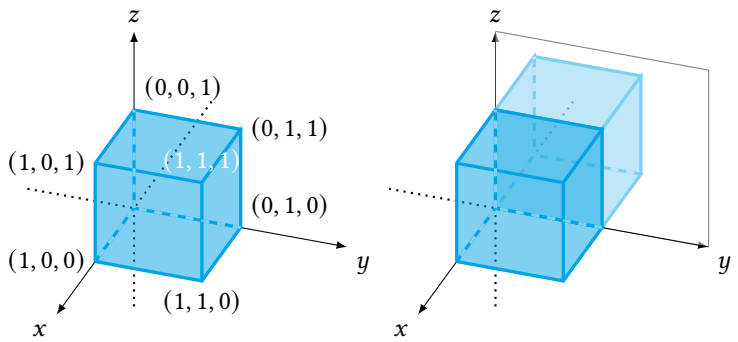


$$\mathbf{E} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$



Question 6 [10 marks]

Consider a unit cube situated in the first octant (with x, y, z all positive) of three-dimensional space, as shown below.



If we place a mirror along the yz-plane, the cube gets reflected into another cube in the second octant (with x negative and y, z positive).

(a) Let T represent the transformation the cube undergoes as it is reflected through the mirror on the yz-plane. For each of the cube's vertices, write down the coordinates of their reflection in the mirror.

(b) Hence, or otherwise, find the standard matrix $\mathbf{T}$ of this transformation.

(c) Is the matrix $\mathbf{T}$ invertible? If so, state $\mathbf{T}^{-1}$ and explain the geometric significance of your answer; otherwise, explain why $\mathbf{T}^{-1}$ does not exist.

(a) Only the x-coordinate of each point changes:

$$\begin{aligned} T(1,0,1) &= (-1,0,1), & T(0,0,0) &= (0,0,0), \\ T(1,1,1) &= (-1,1,1), & T(0,1,1) &= (0,1,1), \\ T(1,0,0) &= (-1,0,0), & T(0,0,1) &= (0,0,1), \\ T(1,1,0) &= (-1,1,0), & T(0,1,0) &= (0,1,0). \end{aligned}$$

(b) The columns of the standard matrix are simply the images of the standard basis vectors $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$:

$$\mathbf{T} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(c) $\mathbf{T}$ is invertible, and $\mathbf{T}^{-1} = \mathbf{T}$: geometrically, this means that a reflection about the yz-plane is undone by a reflection about the yz-plane as well.

Question 7 [10 marks]

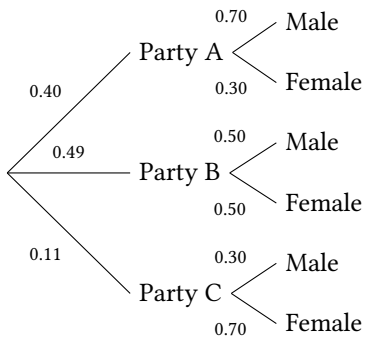
In an election, 40% of all voters voted for Party A, 49% voted for Party B, and 11% voted for Party C. Among those who voted for Party A, 30% were female; among those who voted for Party B, 50% were female, and among those who voted for Party C, 70% were female.

(a) Find the probability that the voter is a female.

(b) If a voter is female, what is the probability that she voted for Party A?

(c) If a voter is female, what is the probability that she voted for Party B or Party C?

We can construct a tree diagram to capture the data:



$P(A \cap M) = 0.40 \cdot 0.70$

$P(A \cap F) = 0.40 \cdot 0.30$

$P(B \cap M) = 0.49 \cdot 0.50$

$P(B \cap F) = 0.49 \cdot 0.50$

$P(C \cap M) = 0.11 \cdot 0.30$

$P(C \cap F) = 0.11 \cdot 0.70$

(a) Since the voting groups are mutually exclusive, we have

$$\begin{aligned} P(F) &= P(A \cap F) + P(B \cap F) + P(C \cap F) \\ &= 0.40 \cdot 0.30 + 0.49 \cdot 0.50 + 0.11 \cdot 0.70. \end{aligned}$$

(b) We compute the conditional probability:

$$P(A \mid F) = \frac{P(A \cap F)}{P(F)} = \frac{0.4 \cdot 0.3}{0.40 \cdot 0.30 + 0.49 \cdot 0.50 + 0.11 \cdot 0.70}.$$

(c) Since there are only three voting groups,

$$P((B \cup C) \mid F) = 1 - P(A \mid F) = 1 - \frac{0.4 \cdot 0.3}{0.40 \cdot 0.30 + 0.49 \cdot 0.50 + 0.11 \cdot 0.70}.$$

Alternatively, since the voting groups are mutually exclusive, we have

$$\begin{aligned} P((B \cup C) \mid F) = P(B \mid F) + P(C \mid F) &= \frac{P(B \cap F)}{P(F)} + \frac{P(C \cap F)}{P(F)} \\ &= \frac{0.49 \cdot 0.5 + 0.11 \cdot 0.70}{0.40 \cdot 0.30 + 0.49 \cdot 0.50 + 0.11 \cdot 0.70}. \end{aligned}$$

Question 8 [10 marks]

Let $X \sim \text{Geom}(p)$ be a geometrically distributed random variable.

(a) Write down the possible values of X.

(b) The geometric distribution is *memoryless*. Briefly explain what this means.

(c) Suppose that $p = 0.2$. Compute $\mathbb{E}(X \mid X \geq 2)$.

(a) X can take on any positive integer: $X = 1, 2, \ldots$.

(b) No matter how many failures have occurred, the distribution of the number of remaining trials until a success is observed is the same. In other words,

$$P(X \geq x + k \mid X \geq k) = P(X \geq x).$$

(c) By the definition of the expected value,

$$\mathbb{E}(X \mid X \geq 2) = \sum_{x=1}^{\infty} x \cdot P(X = x \mid X \geq 2).$$

Note that $P(X = x \mid X \geq 2) = 0$ when $x < 2$. Thus, it suffices to consider the terms when $x \geq 2$: in this case, since X is memoryless,

$$P(X = x \mid X \geq 2) = P(X = x - 1) \text{ for all } x \geq 2.$$

Thus,

$$\begin{aligned}
\mathbb{E}(X \mid X \geq 2) &= \sum_{x=2}^{\infty} x \cdot P(X = x - 1) \\
&= \sum_{x=2}^{\infty} (1 + (x - 1)) \cdot P(X = x - 1) \\
&= \sum_{x=1}^{\infty} P(X = x) + \sum_{x=1}^{\infty} x \cdot P(X = x) \\
&= 1 + \mathbb{E}[X] = 1 + \frac{1}{0.2} = 6.
\end{aligned}$$

Question 9 [10 marks]

A line segment of unit length is cut into two pieces at a randomly chosen point along the segment. Let X be difference in the lengths between the two resulting segments.

(a) Find the probability that $X < 0.2$.

(b) Find the expected value and the variance of X.

Let Y be the length of the longer segment, such that the length of the shorter segment is $1 - Y$. Since the point at which the segment is cut is chosen at random, we have $Y \sim U(0.5, 1)$. Then,

$$X = Y - (1 - Y) = 2Y - 1.$$

(a) We find the probability that

$$X < 0.2 \iff 2Y - 1 < 0.2 \iff Y < 0.6.$$

Thus,

$$P(X < 0.2) = P(Y < 0.6) = \frac{0.6 - 0.5}{1 - 0.5} = 0.2.$$

(b) Since Y is uniformly distributed, we have

$$\mathbb{E}[Y] = \frac{0.5 + 1}{2} = \frac{3}{4}, \quad \mathrm{Var}(Y) = \frac{(1 - 0.5)^2}{12} = \frac{1}{48}.$$

We can thus find the expected value and variance of X:

$$\mathbb{E}[X] = \mathbb{E}[2Y - 1] = 2\mathbb{E}[Y] - 1 = 2\left(\frac{0.5 + 1}{2}\right) - 1 = \frac{1}{2},$$

$$\mathrm{Var}(X) = \mathrm{Var}(2Y - 1) = 2^2\mathrm{Var}(Y) = 4\left(\frac{(1 - 0.5)^2}{12}\right) = \frac{1}{12}.$$

Question 10 [10 marks]

Let X be the time (in hours) before a fully charged mobile device runs out of battery. Suppose that the probability density function of X is given by

$$p(x) = \begin{cases} cx^{-2} & x > 10, \\ 0 & x \leq 10. \end{cases}$$

(a) Find the value of c. Hence, find the cumuluative distribution function of X.

(b) Find the probability that a fully charged device will last more than 20 hours.

(c) Suppose you have six such devices. What is the probability that at least three of them will function for at least 15 hours? State any assumptions.

(a) Since $p(x)$ is a probability density function, we have

$$1 = \int_{-\infty}^{\infty} p(x)\, dx = \int_{10}^{\infty} \frac{c}{x^2}\, dx = \left[-\frac{c}{x}\right]_{10}^{\infty} = \frac{c}{10} \quad \Longrightarrow \quad c = 10.$$

When $x > 10$, the cumulative distribution function is given by

$$F(x) = \int_{10}^{x} \frac{10}{t^2}\, dt = \left[-\frac{10}{t}\right]_{10}^{x} = -\frac{10}{x} - \left(-\frac{10}{10}\right) = 1 - \frac{10}{x}.$$

Thus,

$$F(x) = \begin{cases} 1 - 10x^{-1} & x > 10, \\ 0 & x \leq 10. \end{cases}$$

(b) We compute

$$\begin{aligned} P(X > 20) &= 1 - P(X \leq 20) = 1 - F(20) \\ &= 1 - \left(1 - \frac{10}{20}\right) = \frac{10}{20} = 0.5. \end{aligned}$$

(c) The probability that a single device will function for at least 15 hours is given by

$$\begin{aligned} P(X \geq 15) &= 1 - P(X < 15) = 1 - F(15) \\ &= 1 - \left(1 - \frac{10}{15}\right) = \frac{10}{15} = \frac{2}{3}. \end{aligned}$$

Let Y be the number of devices that will function for at least 15 hours. Assuming that the devices operate independently of one another, we have $Y \sim \text{Binom}(6, 2/3)$. We thus compute

$$\begin{aligned} P(Y \geq 3) &= 1 - P(Y \leq 2) \\ &= 1 - (P(Y = 0) + P(Y = 1) + P(Y = 2)) \\ &= 1 - \left[\binom{6}{0}\left(\frac{2}{3}\right)^0\left(\frac{1}{3}\right)^6 + \binom{6}{1}\left(\frac{2}{3}\right)^1\left(\frac{1}{3}\right)^5 + \binom{6}{2}\left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right)^4\right] \\ &= 1 - \left[1 \cdot \frac{1}{3^6} + 6 \cdot \frac{2}{3^6} + 15 \cdot \frac{4}{3^6}\right] = \frac{656}{729}. \end{aligned}$$

You've made it to the end!